

Panel Data

ECON20222 - Lecture 6

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Dealing with endogeneity

There are 4 main techniques for dealing with endogeneity:

- The First Differenced estimator (FD);
- Differences-in-differences (DiD), as per your projects;
- Instrumental Variables (IV) and
- Regression Discontinuity (RD) [not in the syllabus]
- We will also mention RCTs (see Demo Class 1)

This lecture introduces the First Differenced estimator.

The key requirement is **Panel Data**. We explain using the simplest case, where $T = 2$.

Two-period panel data

Comprise N individuals (or other micro unit) observed exactly $T = 2$ times.

i	t	y_{it}	1	d_t^1	d_t^2	x_{it}	a_i	u_{it}
1	1	y_{11}	1	1	0	x_{11}	a_1	u_{11}
1	2	y_{12}	1	0	1	x_{12}	a_1	u_{12}
2	1	y_{21}	1	1	0	x_{21}	a_2	u_{21}
2	2	y_{22}	1	0	1	x_{22}	a_2	u_{22}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	1	y_{N1}	1	1	0	x_{N1}	a_N	u_{N1}
N	2	y_{N2}	1	0	1	x_{N2}	a_N	u_{N2}

Two-period panel data

Variables in the data:

- y_{it} and x_{it} ; both vary over both $i = 1, \dots, N$ and $t = 1, 2$.
- d_t^1 is a period 1 time dummy; it only varies over t .
- d_t^2 is a period 2 time dummy; it only varies over t .
- 1 is the constant.

Variables not observed:

- a_i does not vary over t . It is a so-called fixed effect, or unobserved heterogeneity;
- u_{it} is an idiosyncratic error. It is assumed to be uncorrelated with everything (like most error terms).

In particular:

$$\text{Cov}(x_{i1}, u_{i1}) = \text{Cov}(x_{i2}, u_{i2}) = 0 \quad (\text{AFD1})$$

Two-period panel data

- The omitted variable a_i is correlated with x_{it} . Eg ability is correlated with education. If not, the OVB formula says OLS is unbiased.
- The crucial assumption is that a_i is constant over time.

The sample version of the model is specified as having 2 errors, one fixed over time, the other time-varying:

$$y_{it} = \beta_0 + \delta_0 d_t^2 + \beta_1 x_{it} + a_i + u_{it} \quad t = 1, 2$$

- This is a so-called **fixed effects** model. - Because of the dummy-variable trap, one of two time dummies must be dropped, here d_t^1 . (Recall one cannot have a constant, male dummy and a female dummy in the same regression.)

Differencing out a_i

We could estimate by OLS, treating $a_i + u_{it}$ as a single composite error. This estimator is called Pooled OLS (POLS).

But $\text{Cov}(x_{it}, a_i) \neq 0$. From the OVB framework, Pooled OLS is biased. Often called heterogeneity bias.

With panel data, we can difference-out the unobserved fixed effect a_i :

$$\begin{aligned}y_{i1} &= \beta_0 + \beta_1 x_{i1} + a_i + u_{i1} \quad (t = 1) \\y_{i2} &= \beta_0 + \delta_0 + \beta_1 x_{i2} + a_i + u_{i2} \quad (t = 2)\end{aligned}$$

Subtracting,

$$\Delta y_i = \delta_0 + \beta_1 \Delta x_i + \Delta u_i.$$

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$$\Delta y_i = \delta_0 + \beta_1 \Delta x_i + \Delta u_i.$$

where:

- $\Delta y_i \equiv y_{i2} - y_{i1}$ is the difference of y_i .
- $\Delta x_i \equiv x_{i2} - x_{i1}$. (In a multiple regression all the regressors are differenced.)
- β_0 and $\beta_0 + \delta_0$ are “macro” effects in that they affect all individuals identically, in the same time period. But only δ_0 is identified. β_0 drops out because

$$\Delta 1 = 0 = \Delta d_t^1 + \Delta d_t^2$$

- δ_0 is the parameter on Δd_t^2 , which is the new regression constant.

Does $\Delta u_i \equiv u_{i2} - u_{i1}$ still has classical properties?

To see what is happening, look at the schematic data above, and subtract the first row from the second for each $i \dots$

Data Structure

$$\Delta y_i = \delta_0 + \beta_1 \Delta x_i + \Delta u_i.$$

	t	y_{it}	1	d_t^1	d_t^2	x_{it}	Δy_{it}	$\Delta 1$	Δd_t^1	Δd_t^2	Δx_{it}
1	1	y_{11}	1	1	0	x_{11}	-	.	-	-	.
1	2	y_{12}	1	0	1	x_{12}	Δy_1	0	-1	1	Δx_1
2	1	y_{21}	1	1	0	x_{21}	.	-	-	..	.
2	2	y_{22}	1	0	1	x_{22}	Δy_2	0	-1	1	Δx_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	1	y_{N1}	1	1	0	x_{N1}	-	.	-	-	-
N	2	y_{N2}	1	0	1	x_{N2}	Δy_N	0	-1	1	Δx_N

The change in y_{it} between periods 1 and 2 is regressed on the change in the regressor(s) and a constant, using N obs.

First-difference estimator (FD)

- The variables are not defined for period 1 and so the dataset comprises just N obs. Depending on software, the investigator needs to be careful about organisation of the data to compute the correct change. (See code file)
- OLS on the differenced specification is called the first-difference estimator (FD).
- In this example, it is a Simple Regression. In general, FD is a multiple regression where all variables are differenced.
- The important issue is whether the same assumptions for OLS are satisfied? If yes, FD is unbiased.

The assumptions are:

- ① The RHS variable is exogenous (uncorrelated with the error);
- ② The RHS variable cannot be constant.

Exogeneity

Is Δx_i uncorrelated with Δu_i ?

$$\begin{aligned}\text{Cov}(\Delta x_i, \Delta u_i) &= \text{Cov}[(x_{i2} - x_{i1}), (u_{i2} - u_{i1})] = \\ &\text{Cov}(x_{i2}, u_{i2}) + \text{Cov}(x_{i1}, u_{i1}) - \text{Cov}(x_{i1}, u_{i2}) - \text{Cov}(x_{i2}, u_{i1})\end{aligned}$$

- All 4 covariances need to be zero. We have already assumed the first pair are zero (AFD1). If AFD1 is true, there is no reason why the second pair (“across” $t = 1, 2$) should not be zero either.
- To summarise, if x and u are uncorrelated, then so are Δx and Δu .

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Variation in Δx_i

- Each Δx_i must have some variation across i . Age, gender, and education are good examples of Δx_i being constant. Iow, some issues (education, discrimination etc) cannot be investigated using FD.
- Even if Δx_i is not constant, it might not vary much, leading to large standard errors.
- Measurement error gets worse with differencing. We know from Lecture 4 that this means FD gets more biased.

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Variation in Δx_i

- See what happens with male dummy m , age a and education e :

i	t	y_{it}	d_t^2	m	a	e	Δy_{it}	Δd_t^2	Δm	Δa	$\Delta e?$
1	1	y_{11}	0	1	18	11	.	.	.	.	
1	2	y_{12}	1	1	24	11	Δy_1	1	0	6	
2	1	y_{21}	0	0	28	13	.	.	.	.	
2	2	y_{22}	1	0	34	13	Δy_2	1	0	6	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
N	1	y_{N1}	0	1	51	16	.	.	.	.	
N	2	y_{N2}	1	1	57	16	Δy_N	1	0	6	

The Wage Curve

It has become a stylised fact that the elasticity of real wages with respect to the unemployment rate is -0.10 . Danny Blanchflower and Andrew Oswald have written extensively on this issue, generating much supporting evidence.

In what follows, I use Understanding Society Waves 1 and 3.

- There are 12 regions in the UK.
- Each wave covers 3 years, so there are 5 years. The unemployment rate varies by year and region (60 distinct values minus NI/2013).
- The next slides tabulate log average hourly pay and the unemployment rate, by region.
- Finally, we compare POLS and FD estimates.

Data in waves

```
##           waves
## n_waves      1      3
##           1  9666  4112
##           2 13078 13078
```

Some Summary Statistics

```
##
## =====
## Statistic      N          Mean      St. Dev.    Min      Max
## -----
## lnhrpay      39,934      2.284      0.635     -7.816    8.868
## urate        39,934      7.877      1.303      5.800   10.800
## year         39,934  2,010.393      1.146      2,009   2,013
## Dlnhrpay     13,078     -0.009      0.524     -10.381    9.522
## Dlnurate     13,078      0.037      0.065      -0.464    0.547
## -----
```

Statistics by region

```
## # A tibble: 12 x 4
```

##	region	n	mean_lnhrrpay	mean_urate
##	<fct>	<int>	<dbl>	<dbl>
##	1 north east	1576	2.21	9.88
##	2 north west	4280	2.24	8.44
##	3 yorkshire and the humber	3247	2.20	9.00
##	4 east midlands	3107	2.20	9.15
##	5 west midlands	3454	2.23	7.67
##	6 east of england	3724	2.32	6.58
##	7 london	5736	2.42	9.30
##	8 south east	5125	2.39	6.10
##	9 south west	3119	2.25	6.11
##	10 wales	1831	2.14	8.45
##	11 scotland	3020	2.27	7.76
##	12 northern ireland	1715	2.24	6.78

Comments on the data

- 13,078 individuals are observed twice, 9,666 + 4,112 only once.
- There are 39,934 individual/waves in total.
- The 13,778 individuals observed once do not contribute to the FD estimator. This leaves 13,078 who do contribute (26,156 observations)
- Some unemployment rates are twice others.
- London has the highest pay and second highest unemployment rate
- Unemployment changes because of regional (macro) variations over time (2009-2013) and because some individuals move region.

Estimate Pooled OLS

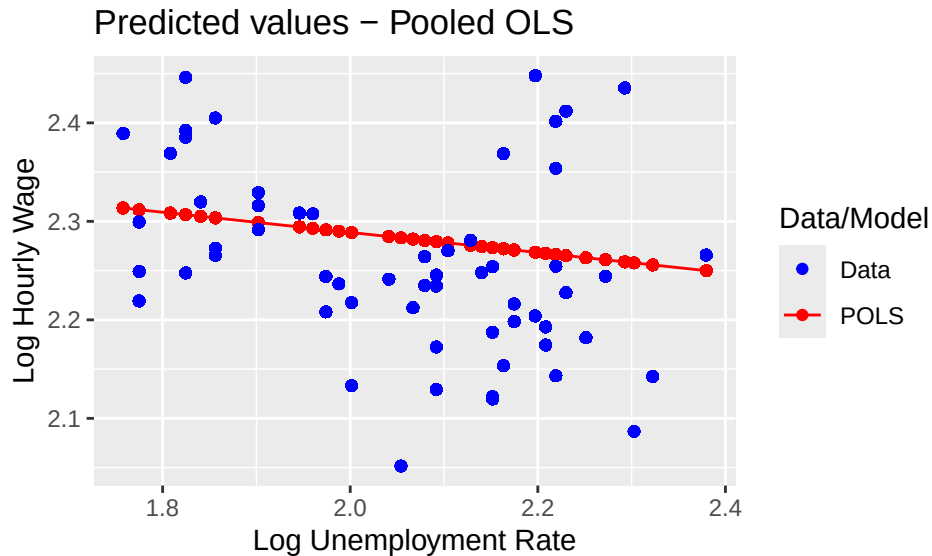
```
POLS0 <- lm(lnhrpay~lnurate, data = pdata_USoc)
```

```
# or using the plm function
```

```
POLS0a <- plm(lnhrpay~lnurate, data = pdata_USoc, model = "pooling")
```

```
pdata_USoc$pred_POLS0 <- POLS0$fitted.values
```

Predicted values - Pooled OLS



Different Models

```
##
## =====
##                               Dependent variable:
##                               -----
##                               lnhrpay
##                               (1)      (2)      (3)      (4)
## -----
## lnurate      -0.102***   -0.097***   -0.096***   0.042
##              (0.022)     (0.022)     (0.029)   (0.085)
##
## wave3              -0.019***   -0.005
##              (0.005)     (0.005)
##
## Constant      2.493***   2.491***   2.544***   -0.011*
##              (0.046)     (0.046)     (0.058)   (0.006)
##
## -----
## Observations   39,934     39,934     26,156     13,078
## R2             0.001      0.001      0.001      0.00003
## Adjusted R2    0.001      0.001      0.001     -0.00005
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
##              Robust standard errors in parenthesis
```

Comments on the estimation}

Using POLS, col(3), the wage-curve elasticity is exactly as Blanchflower and Oswald found (-0.1):

$$\frac{\% \Delta \hat{y}_i}{\% \Delta x_i} = -0.096(0.029)$$

The correct way to interpret the POLS elasticity is the effect of, say, a 10% increase in the unemployment rate (say from 8% to 8.8%) on real hourly pay, namely

$$\% \Delta \hat{y}_i = -0.096 \% \Delta x_i = -0.096 * 0.10 = -0.0096$$

that is, a drop of 0.0096 log-points (1 percent).

The FD estimate 0.042 is half the size, changes sign, and with a larger standard error (0.060), it is insignificant. It is interpreted in exactly the same way. The difference is that we have now controlled for any time-invariant heterogeneity a_i .

You must **not** interpret this as $\frac{\% \Delta (\Delta \hat{y}_i)}{\% \Delta (\Delta x_i)} = 0.042$.

Comments on the estimation

- We can interpret a_i as some unobserved time-invariant variable that means that some individuals are attracted to higher paying regions (such as London).
- Because education is omitted in this kind of analysis, a_i could be education and ability/motivation.
- Comparing (3) and (4), and using the OVB formula (Lecture 5), we can work out the sign of $\text{Corr}(x_{it}, a_i)$:

$$\log y_{it} = -0.005d_t^2 - 0.096x_{rt} + \text{const} \quad (\text{POLS})$$

$$\log y_{it} = -0.011d_t^2 + 0.042x_{rt} + a_i \quad (\text{FD})$$

where

$$a_i = \delta_0 + \delta_1 x_{rt} + \text{error} .$$

... continued

Using $\tilde{\beta}_1 - \hat{\beta}_1 = \hat{\delta}_1$ ($\beta_2 = 1$), $\hat{\delta}_1 = -0.138$, and so

$$\hat{a}_i = \hat{\delta}_0 - 0.138x_{rt}$$

- So, in high (log) unemployment rate regions, a_i is lower (measured in the same units as the dep variable). This makes good sense.
- There is lots of variation in Δx_i (Dlnurate). See the summary of D variables.
- Is measurement error an issue? Unambiguously No. The interviewers know exactly which region households are located. We have “matched” the official ONS unemployment rates to that region variable.
- Finally, FD estimation can be done on more than just $T = 2$ waves. One then ends up with a panel of differenced data.

Concluding comments

- Heterogeneity bias comes about because of $\text{Corr}(x, a) \neq 0$.
- With panel data, is the unobserved heterogeneity time-invariant? Then the problem is written $\text{Corr}(x_{it}, a_i) \neq 0$.
- If so, the a_i can be differenced away, leaving Δy regressed on Δx s using OLS.
- The estimator is called FD.
- The idea is best explained with $T = 2$, but extends to $T > 2$.
- There are other panel data estimators that are based on the same set-up, notably so-called fixed effects (FE).
- The price paid for these estimators is that there may be zero or little variation in Δx .
- We next deal with endogeneity by extending the FD estimator to the so-called difference-in-difference estimator. It is motivated by the set-up in this lecture...